ACADEMIC & RESEARCH PORTFOLIO

Rifat Hossain Talukder

Transportation and urban systems

Climate resilience and spatial vulnerability

GIS, remote sensing and geospatial modelling

Research across infrastructure-led mobility, land-use change, environmental exposure and access to essential services.



PADMA BRIDGE REGION

Dhaka, Bangladesh rifatht21@gmail.com Khulna University

SELECTED EVIDENCE

01 / TRANSPORT

Counterfactual analysis

Observed-versus-no-bridge traffic, behavioural evidence and scenario uncertainty.

02 / CLIMATE

Random Forest modelling

Landsat indicators and cyclone-related flood vulnerability in coastal Bangladesh.

03 / SPATIAL

Land-use change

Remote sensing, Markov transitions and Cellular Automata scenarios.

ACADEMIC SNAPSHOT

Bachelor of Urban and Rural Planning · Khulna University (2023)

Undergraduate thesis · Padma Bridge urban-growth scenarios to 2046

Research experience · University Grants Commission of Bangladesh, UNDP, CEGIS and field research

8.0

IELTS overall
October 2024

SELECTED RESEARCH · PUBLICATIONS · EXPERIENCE · METHODS

RESEARCH PROFILE

Questions, methods and evidence

Planning research that connects infrastructure, environmental change and spatial inequality.

RESEARCH FOCUS

My work examines how new infrastructure and changing environmental conditions affect travel, settlement patterns, physical exposure and access. I use spatial data to identify change, quantitative models to evaluate possible mechanisms, and field or archival evidence to interpret results.

01

Transport & access

Induced travel, route choice, network conditions, urban growth and mobility equity.

02

Climate resilience

Flood and cyclone vulnerability, infrastructure exposure and resilient service access.

03

Geospatial methods

Earth observation, land-use modelling, GIS, Random Forest and mixed-methods research.

RESEARCH PRACTICE

SPATIAL & QUANTITATIVE

- Landsat imagery, image-derived indices, raster/vector processing and GIS mapping.
- Random Forest regression; R^2, MAE and RMSE-based evaluation.
- Counterfactual reasoning, Monte Carlo uncertainty analysis and scenario forecasts.

FIELD & INTERPRETIVE

- Interviews, focus groups and key-informant interviews.
- Thematic coding, literature synthesis and research reporting.
- Archival maps, planning documents, built-environment history and spatial reconstruction.

CURRENT RESEARCH

Induced traffic of the Padma Multipurpose Bridge: mixed-methods counterfactual study supervised by Professor Dr. Md Ahsanul Kabir.
Additional lines of work: historical-spatial research on the University of Dhaka campus and an extension of the coastal cyclone-vulnerability modelling workflow.

SELECTED RESEARCH / 01

Traffic after the Padma Bridge

Deciphering Traffic Dynamics: Analysing the Induced Traffic of the Padma Multipurpose Bridge · ongoing

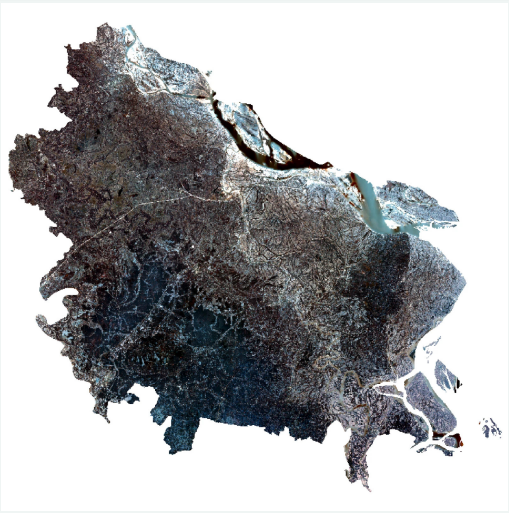
CORE QUESTION

How much of the observed cross-Padma traffic exceeds a plausible no-bridge baseline, and what part of that difference may reflect new travel rather than route diversion or mode transfer?

20,240
VEHICLES / DAY
Observed bridge traffic

8,052
VEHICLES / DAY
Preferred no-bridge estimate

12,188
VEHICLES / DAY
Bridge-attributable excess



Study geography: Padma Bridge corridor and surrounding land cover.

STUDY DESIGN

01 Construct a pre-bridge movement baseline.

02 Estimate no-bridge traffic with temporal and spatial comparisons.

03 Use interviews to distinguish possible travel mechanisms.

04 Propagate uncertainty through Monte Carlo simulations.

FINDINGS & INTERPRETATION

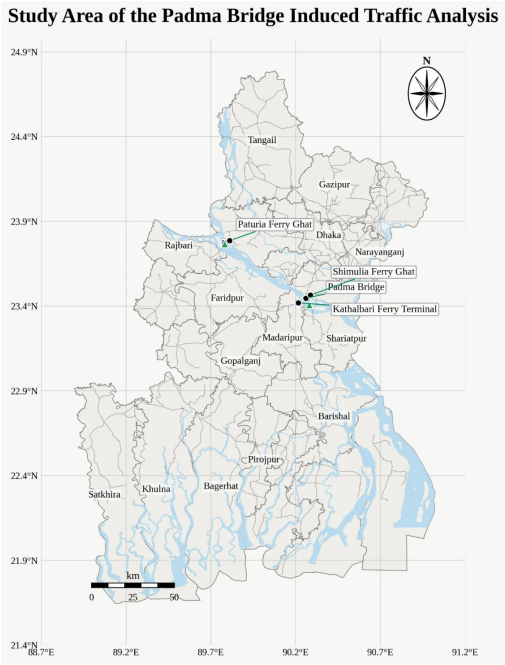
60.2% of observed traffic is represented by the main excess estimate. A Monte Carlo simulation yielded a median excess of **11,696 vehicles/day**. Interviews indicated additional trip frequency and newly feasible journeys alongside diverted routes, destination changes and mode transfer.

Interpretive boundary. Bridge-attributable excess traffic is not equivalent to pure induced demand. The no-bridge condition is counterfactual, and the estimates depend on historical data and model assumptions. The 2054 range of 33,169–44,537 vehicles/day is scenario-based.

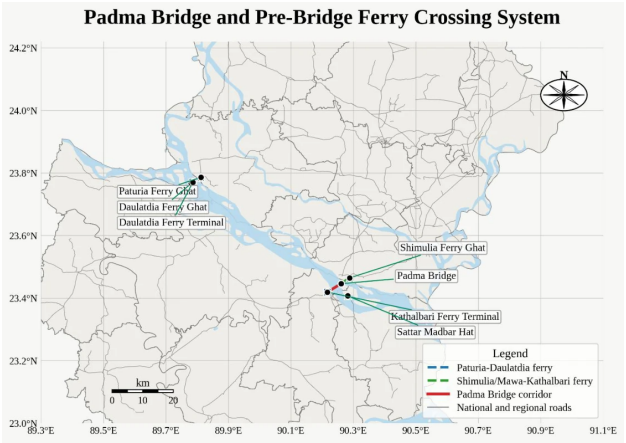
SELECTED RESEARCH / 01 / VISUAL EVIDENCE

Following the traffic change

Padma Bridge corridor, historical crossing context and sensitivity of the traffic estimates.



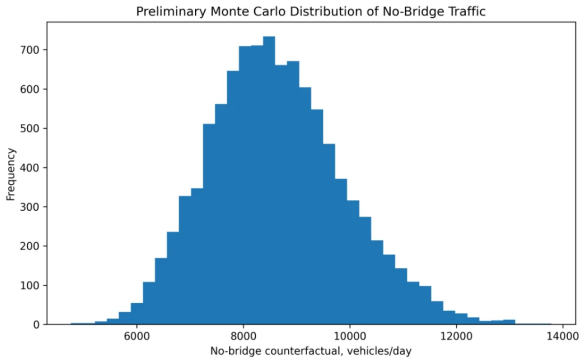
Padma Bridge study geography and transport corridor. Source: ongoing mixed-methods study.



INTERPRETATION

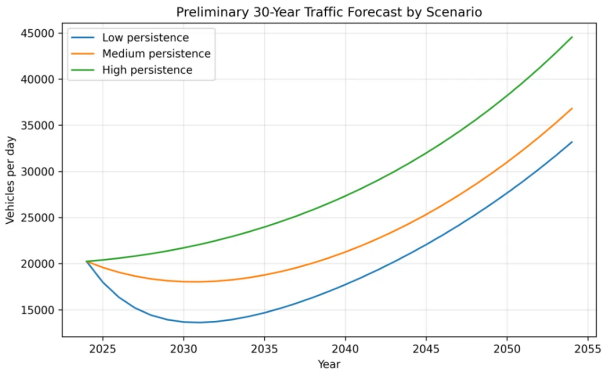
Earlier ferry and launch routes are part of the reference condition. Some bridge traffic represents diversion and mode transfer rather than newly generated trips.

COUNTERFACTUAL UNCERTAINTY



Monte Carlo estimates account for uncertainty in the no-bridge traffic baseline.

TRAFFIC SCENARIOS THROUGH 2054



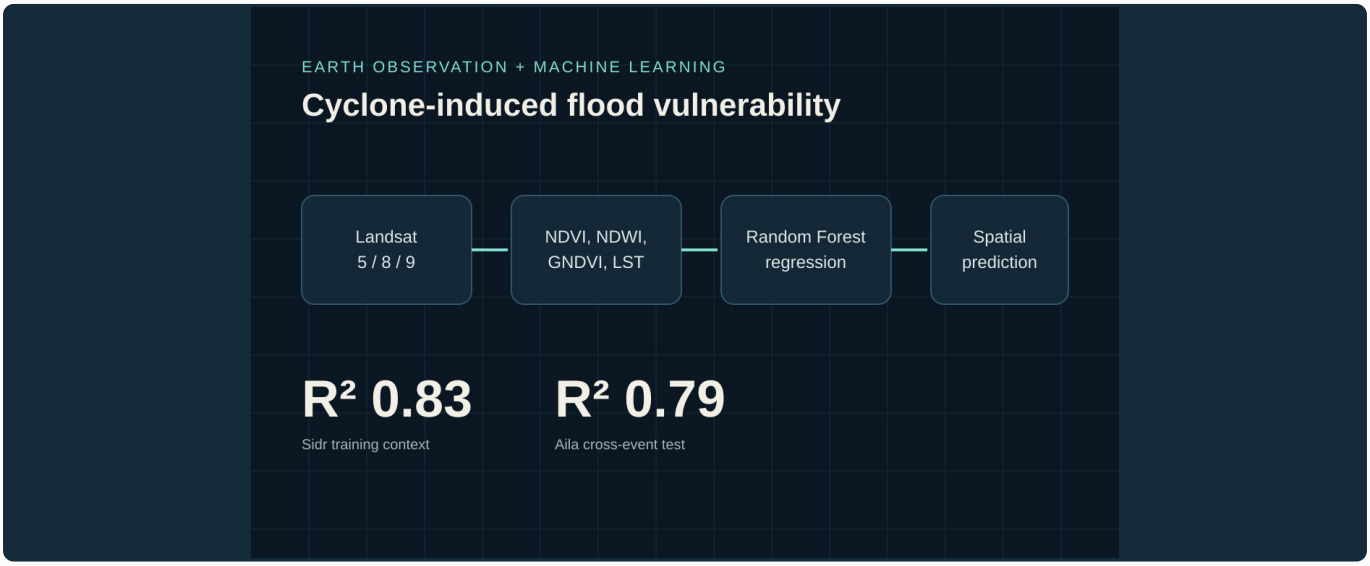
Long-term forecasts are conditional scenarios, not observed future traffic.

SELECTED RESEARCH / 02

Cyclone-related flood vulnerability

UNDP Sundarbans Area Based Programme · Khulna, Satkhira and Bagerhat · completed 2023

Developed a remote-sensing and machine-learning workflow to model spatial variation in cyclone-related flooding using Landsat-derived environmental indicators and Random Forest regression.



Model workflow and reported evaluation metrics from the study.

0.83

SIDR SCENARIO · R^2

0.79

AILA SCENARIO · R^2

INPUTS & METHODS

Landsat 5, 8 and 9 imagery; GNDVI, NDVI, NDWI, land-surface temperature, salinity index and simple-ratio measures. Temporal NDWI change served as a flood-intensity proxy; Random Forest regression captured nonlinear relationships.

MODEL EVALUATION & LIMITS

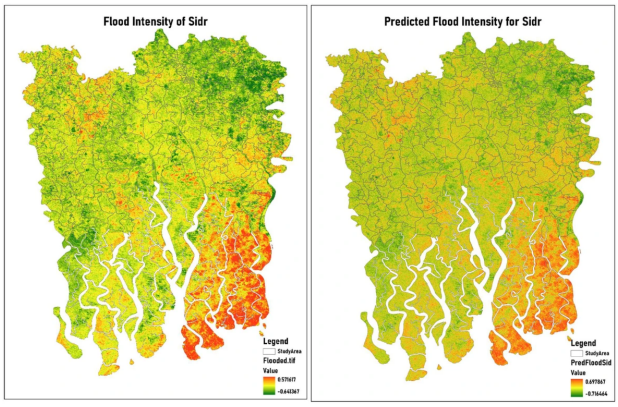
Reported scenario R^2 values were 0.83 (Sidr) and 0.79 (Aila), alongside MAE and RMSE assessment. Interpretation is limited by event coverage, differences among satellite acquisitions and the use of remote-sensing proxies for observed flood conditions.

SELECTED RESEARCH / 02 / MAPPED RESULTS

Cyclone-related flood mapping

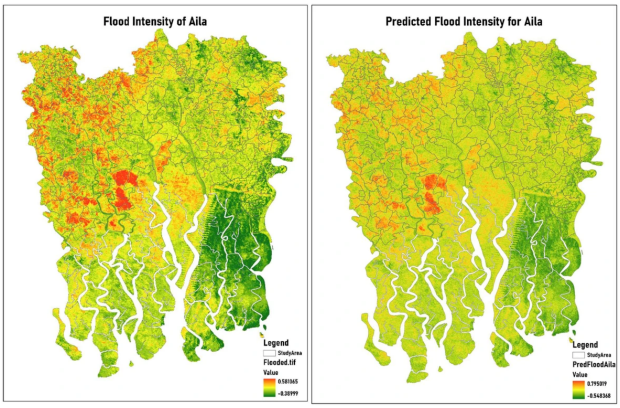
Original evaluation and scenario maps from the UNDP Sundarbans-area research report.

CYCLONE SIDR



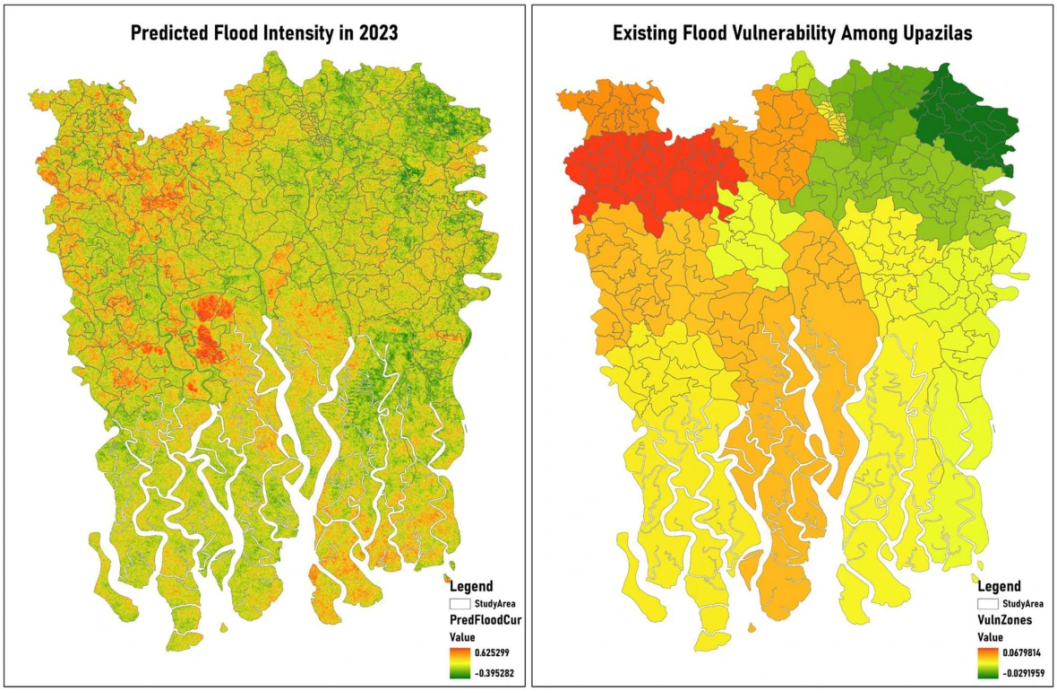
Flood-intensity proxy: reference and Random Forest prediction. Reported R^2: 0.83.

CYCLONE AILA



Flood-intensity proxy: reference and Random Forest prediction. Reported R^2: 0.79.

MAPPED CONTEMPORARY VULNERABILITY



Scenario output for Khulna, Satkhira and Bagerhat using satellite-derived indices. These maps are proxies for flood-related conditions rather than direct observations of total disaster loss.

SELECTED RESEARCH / 03

Urban-growth scenarios near the Padma Bridge

Bachelor of Urban and Rural Planning thesis · Khulna University · February 2023

RESEARCH QUESTION

How might fixed cross-river connectivity reshape built-up land in the Padma Bridge region? In the absence of a long post-opening record, the Jamuna Bridge region provided an analogue for modelling infrastructure-associated urban growth.



Upazila-level scenario output from the thesis.

MODEL WORKFLOW

01 Tested four classifiers on 10 Landsat scenes; selected Random Forest.

02 Modelled pre/post-Jamuna change through Markov transitions and weights of evidence.

03 Adapted change patterns to Padma using 13 spatial predictors.

04 Ran two 2046 scenarios with Cellular Automata in Dinamica EGO.

DATA & SCENARIO OUTPUT

10
LANDSAT SCENES

13
SPATIAL PREDICTORS

11.87%
MODELLED ADDITIONAL GROWTH

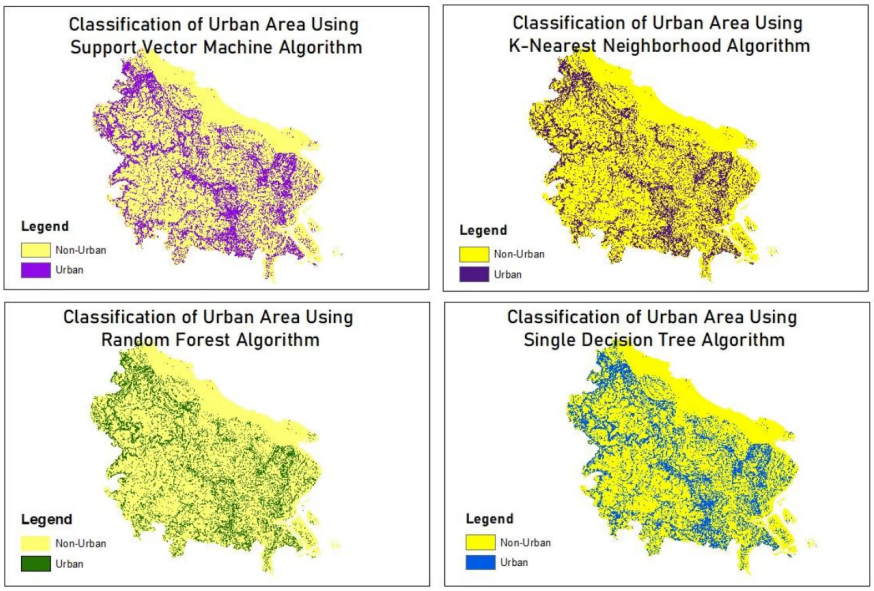
The thesis model estimated approximately 317 km² of additional built-up growth over a 24-year scenario under the specified analogue and transition assumptions. These are conditional planning projections, not a measured causal impact of bridge construction.

SELECTED RESEARCH / 03 / ORIGINAL THESIS FIGURES

How the land-use model was built

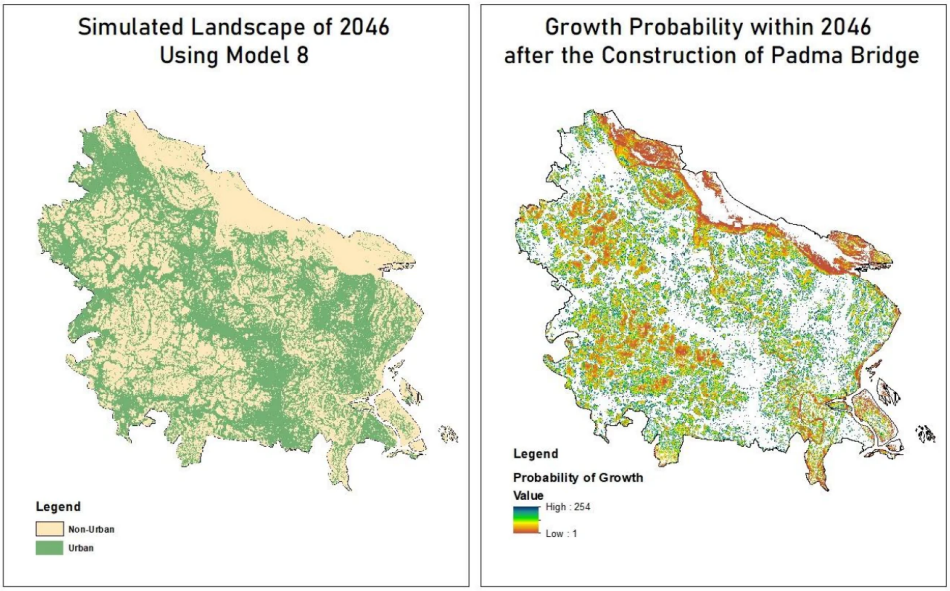
Bachelor of Urban and Rural Planning dissertation, Khulna University, February 2023.

FOUR METHODS COMPARED FOR BUILT-UP LAND CLASSIFICATION



Support Vector Machine, K-Nearest Neighbours, Random Forest and a single decision tree were compared. Random Forest was selected for subsequent land-cover mapping.

2046 SCENARIO OUTPUTS



The model compared an unadjusted growth scenario with an analogue-based scenario drawing on observed land-use change around the Jamuna Bridge. Estimated additional built-up growth: about 317 km² (11.87%).

SCHOLARLY OUTPUT

Published and developing research

Published conference work, a manuscript under review and active research development.

PUBLISHED / CONFERENCE PROCEEDINGS

Assessing Spatial Flood Vulnerability of Greater Rangpur Region in Bangladesh Using GIS: A Tool for Resilience Planning at Regional Scale

Himel, T. I., Talukder, R. H., Hossain, M. Z., Naz, S., Arfin, S., & Fariha, J. N. (2021). In *Planning for Inclusiveness and Sustainability in Post Pandemic Era* (pp. 228–244). Bangladesh Institute of Planners. ISBN 978-984-35-2613-7.

18
INDICATORS

Collaborative GIS assessment combining physical exposure and socioeconomic sensitivity. The indicator framework was normalized and integrated to compare flood-vulnerability patterns across Greater Rangpur. Results indicated differences associated with poverty, agricultural dependence, service gaps, elevation and proximity to rivers.

READ CONFERENCE PROCEEDINGS ↗

MANUSCRIPT / UNDER REVIEW

A Principal Components Analysis (PCA) Based Novel Approach for Assessing Climate Change Induced Loss and Damage in the Coastal Region of Bangladesh

Talukder, R. H., Rahman, M. A., Arfin, S., Fariha, J. N., Husna, A. U., Ha-mim, N. M., Hossain, M. Z., & Rahaman, K. R. Environmental Development (Elsevier) | manuscript under review.

RESEARCH IN DEVELOPMENT

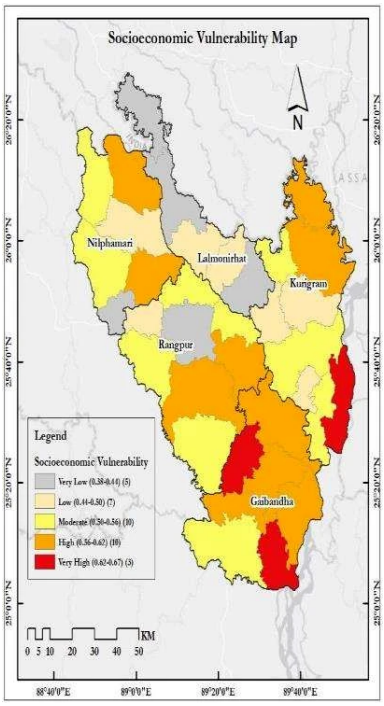
- **Cyclone modelling extension**
  Temporal validation, refreshed remote-sensing inputs and added infrastructure/contextual vulnerability layers.
- **University of Dhaka historical-spatial research**
  Archival maps, planning records, construction histories and spatial reconstruction of campus development, 1921–2018.
- **Climate-resilient service access**
  Concept-stage network GIS work on flood disruption and access to healthcare and other essential services in Dhaka.

PUBLISHED RESEARCH / ORIGINAL MAPS

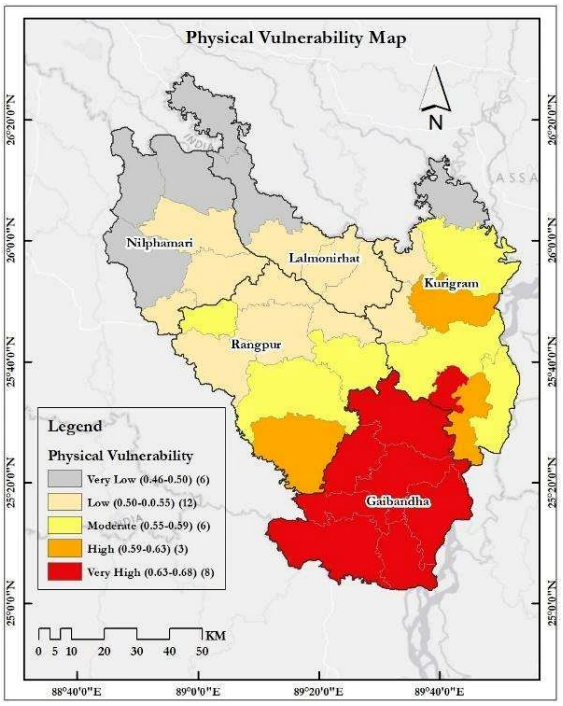
Where flood vulnerability differs

Greater Rangpur regional assessment, Bangladesh Institute of Planners conference proceedings (2021).

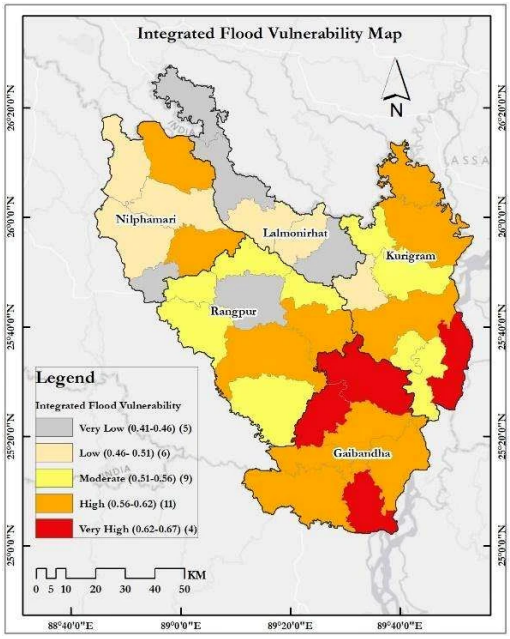
TWO COMPLEMENTARY VULNERABILITY DIMENSIONS



Socioeconomic vulnerability



Physical vulnerability



INTEGRATED FLOOD VULNERABILITY

The study combined 18 indicators covering physical exposure and socioeconomic sensitivity across Gaibandha, Kurigram, Lalmonirhat, Nilphamari and Rangpur.

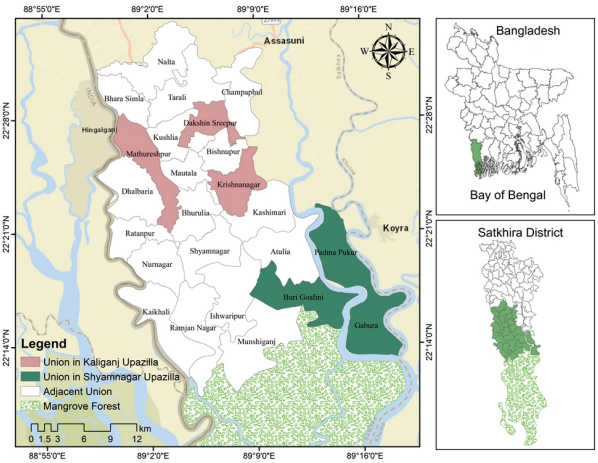
The integrated map identifies regional differences rather than treating all districts as equally exposed. It supports screening where finer local investigation is needed.

RESEARCH MANUSCRIPT / UNDER REVIEW

Measuring coastal loss and damage

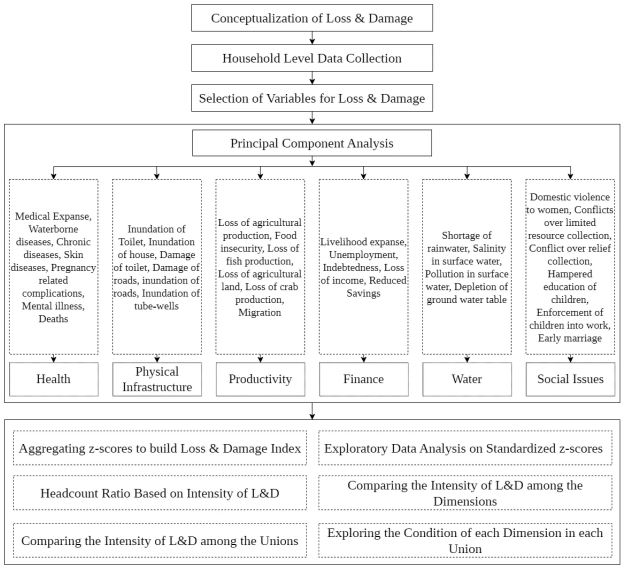
Satkhira household study: principal components analysis and a six-dimension composite index.

STUDY GEOGRAPHY



Selected unions in Shyamnagar and Kaliganj upazilas, Satkhira District.

ANALYTICAL FRAMEWORK



Household indicators, component extraction and comparative assessment.

FROM SURVEY RESPONSES TO AN INDEX

301

households

Primary survey in six coastal unions

6

dimensions

Health, assets, productivity, social, finance and water

PCA

method

Component scores inform an aggregate index

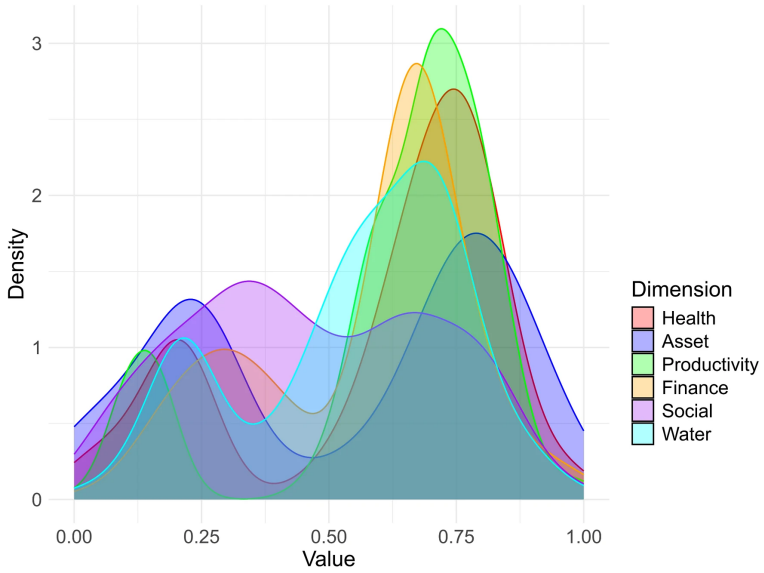
The manuscript also compares indexed outcomes across unions using nonparametric tests. Its findings describe reported losses and damages within the sampled communities.

RESEARCH MANUSCRIPT / MAPPED & STATISTICAL EVIDENCE

Which losses are concentrated?

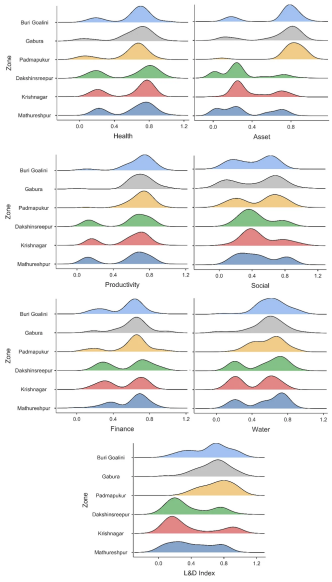
Original statistical figures from the coastal loss-and-damage manuscript.

DISTRIBUTION ACROSS SIX DIMENSIONS



Smoothed distributions of household index scores for health, physical assets, productivity, social, finance and water.

VARIATION ACROSS UNIONS



Original multi-panel density figure.

REPORTED FINDINGS

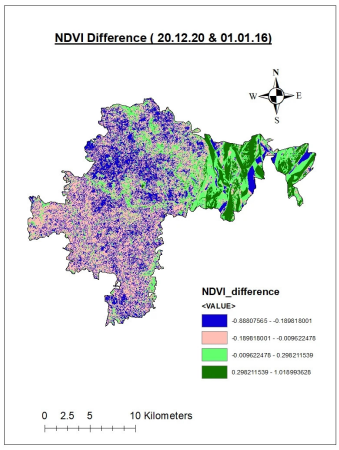
Productivity had the highest reported prevalence of high-impact loss and damage; social impacts were the lowest across the six dimensions.

Burigoalini, Gabura and Padmapukur recorded comparatively greater impacts in the study. The results describe surveyed households and should not be read as a full census of coastal Bangladesh.

ADDITIONAL APPLIED WORK

Range across planning and GIS

Selected coursework and technical studies with spatial, transportation and municipal applications.

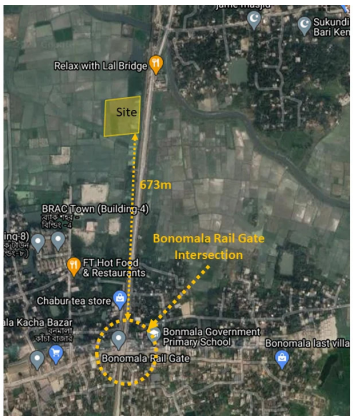


REMOTE SENSING / COURSEWORK

Land-cover change in Gaibandha

Prepared multi-date satellite imagery, land-cover classifications and NDVI products to compare vegetation and land-cover patterns from 2016 to 2020. Outputs included class areas, change maps and comparisons of classification approaches.

Scope: two-date coursework analysis; sensitive to seasonality, atmospheric effects and classification uncertainty.



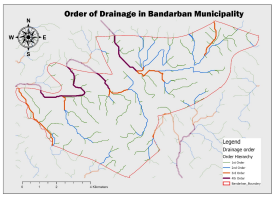
TRANSPORTATION / INDIVIDUAL COURSEWORK

Sukundirbag Tower traffic impact assessment

Recorded mixed-traffic movements, converted counts to passenger-car units, estimated development-related trip generation and examined a five-year background-growth scenario. Proposed access and junction-operation improvements.

Scope: limited survey and simplified capacity assumptions.

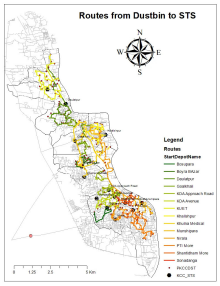
GIS HYDROLOGY



Bandarban drainage & catchments

DEM flow direction, stream order, pour points and catchment delineation.

MUNICIPAL GIS



Solid-waste collection routing

GIS network analysis of collection areas, transfer locations and vehicle routes.

EXPERIENCE & READINESS

Research, teaching and technical practice

Selected appointments and tools supporting graduate research and instructional work.

RESEARCH & TECHNICAL APPOINTMENTS

2021–2024

Research Assistant · University Grants Commission

GIS, land-cover, historical maps and archival evidence for campus development and the evolving built environment of Dhaka.

2023

UNDP research intern · Sundarbans Programme

Designed a Landsat-based Random Forest workflow and interpreted mapped cyclone-related flood vulnerability.

2023

Qualitative Researcher · Tiller Bangladesh

Conducted interviews, focus groups and KIIs; transcribed and coded fieldwork for research reporting.

2021

Technical Consultant · CEGIS

Digitized census-related spatial data and performed positional, scale and consistency checks on GIS layers.

TEACHING & LEADERSHIP

Teaching Assistant
Khulna University · Jan 2024–Jun 2025

Supported GIS, remote sensing, transportation planning and traffic simulation instruction; guided spatial-data preparation and interpretation.

Head & Founding Member
Research and Academic Excellence Department · IELTS School BD
Jul 2025–present

Teach research design, literature review and methods; mentor analytical software use and oversee academic-support workflows.

ACADEMIC CREDENTIALS

Khulna University
Bachelor of Urban and Rural Planning · 2023
CGPA 3.42 / 4.00

SELECTED METHODS & TOOLS

GIS / remote sensing ArcGIS Pro · QGIS · ArcPy · Google Earth Engine · ERDAS IMAGINE · Dinamica EGO

Data / analysis Python · R · SPSS · Jamovi · NVivo · Random Forest · model evaluation · qualitative coding

Transport / spatial Counterfactual analysis · uncertainty modelling · accessibility concepts · TransCAD, MATSim and VISSIM exposure

RESEARCH & GRADUATE ASSISTANTSHIP INQUIRIES rifatht21@gmail.com